User Account Model

Software Requirements / Logical Data Model

1. Purpose

Represents credentials and access identity for people using the Hostel Management System.

2. Responsibilities

• Support authentication for permitted users.

• Associate a person with an authorized role, such as student, hostel administrator or security staff, where those roles are supported by the system.

• Represent account state and security-related metadata without exposing secret credentials.

3. Proposed Logical Attributes

The following is a documentation proposal, not a claim that every field is already implemented. Confirm exact field names, types, constraints and cardinalities against the project ERD/database schema.

Attribute	Meaning / Requirement
account_id	Unique account identifier.
username_or_email	Login identifier, according to the chosen authentication design.
password_hash	Password verifier stored as a secure hash; never store plaintext passwords.
role	Role or permission group assigned to the account.
account_status	Whether the account is enabled, disabled or otherwise restricted.
created_at / updated_at	Timestamps, if maintained by the application.

4. Relationships

• A student account may link to one student profile.

• Staff accounts may be linked to staff records if such a model exists in the full schema.

• Account activity may generate audit-log records.

5. Validation and Business Rules

• Authentication must not reveal whether a password or account detail was the specific cause of failure beyond approved messaging.

• Passwords must be hashed using a suitable password-hashing algorithm.

• Authorization must be checked on protected actions, not only by hiding interface controls.

• Disabled accounts must not gain access to protected functionality.

6. Traceability / Notes

Role names and account fields should be reconciled with the authentication design and ERD.

Status: Draft for review. Align this document with the approved SRS, ERD, class diagram, and institutional policies before treating it as final.